Joel Subach
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OBJECTIVE: Position in Academic or Research Science Focusing on Nano-Biomedical Engineering (Protein or Nucleic Acid Engineering)

OCCUPATIONAL EXPERIENCE:
○ > 25 Years Experience Teaching-Developed/Implemented lesson plans for Biology,
Chemistry, Physics, and Math through lectures and laboratories in Private/Public High Schools and private tutoring (College/University level) in Connecticut and Los Angeles California; 1989-2012

○ 1-2017-5-2017, Manhattan College, Science Instructor (Chemistry, Physics, Earth Science, Biology, Nanotechnology), Riverdale, New York City, New York

EDUCATION:
○ **BA Degree**, Physical Anthropology; Central Connecticut State University; 1990
Honor Roll, CCSU 1989-90; Junior & Senior years of BA GPA 3.0 1989-1990

○ **Post–Graduate**, 1991-2012; Science; UCLA; Post Baccalaureate Science Certificate; GPA 3.315

○ **MS Degree**, Nano-Engineering; NCSU, 2014, Specialization in Biomedical Sciences and Technical Electives in Advanced Mathematics for Scientists and Engineers I and II, GPA 3.5

○ **Ph.D**; Biomedical Informatics/Nano-Medicine; Rutgers University, 2014-2020, New Jersey, GPA 3.964

○ **Spanish Speaking Certified Level B1.1**

○ Experience in Teaching in a Multi-Ethnic Environment, Canvas LMS Savvy

PUBLISHED WORKS:
○ Doctoral Dissertation, Title: Pannexin-1 In Silico Modeling Towards Physiological and Pathological Functioning; Year Published 2020, link below: (Computational Molecular Protein Design of Uncrystallized Protein via Homology Modeling and Ligand-Docking)
https://rucore.libraries.rutgers.edu/rutgers-lib/62581/

COMPUTATIONAL SKILL-SET:

○ **MODELLER:** Homology Protein Modeling; **GalaxyWeb:** Protein Oligomer Structure Prediction; **MedusaDock:** Ligand Receptor Docking; **NCBI, Phyre^2 BLAST:** Template Selector; **Clustal, PROMALS3D, Phyre^2, RaptorX/CNFpred:** Alignment Predictor; **TMpred, TMHMM, SOSUI, RaptorX, Phyre^2:** Topology Predictor; **GROMACS**: Membrane Protein-Ligand Complex Umbrella Sampling; **MATLAB**

Dr. Joel Subach

To whom it may concern,

My current holding of a Ph.D. in Biomedical Informatics and Nanomedicine at Rutgers University will offer your Research Institution an eclectic academic background towards *in silico* molecular design software skills contributing to innovative molecular modeling/simulation research. Computationally my experience encompasses novel 3-D *in silico* protein design and nanoparticle ligand-receptor dynamic simulation dockings. My future endeavors include bringing a novel nano-medical *in silico* engineering design/approach to a lab performing cutting-edge research within my area of interest, nanoparticle-receptor engineering/drug design via molecular dynamic simulations.

I am available for an interview at your convenience, thank you for your time and considerations and may we enjoy each others success.

Kind Regards,

Joel Subach

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Teaching Philosophy

Being Spanish-English bilingual with over 25-years of teaching experience in a diverse multi-ethnic environment coupled to of having lived overseas for several years has matured my teaching skills to be unique, polished, cultured, understanding and matured. This experience connected to my eclectic terminal education beneath the umbrella of Science, Engineering and Math has indeed developed myself into a positive, knowledgable and fair instructor connecting with students to bring out their best academically. My goal as an instructor of both non-science and science majors is to have them successfully complete my courses enlightened and enriched with developed critical thinking and problem solving skills that they may use in their careers, graduate school and life.

Pannexin1 Gating Structural Determination

ABSTRACT

Background:

In silico design and simulation has become an integral methodology towards drug discovery. Pannexin-1 is a ubiquitous expressed Cryo-EM mapped transmembrane ion protein channel controlling a multitude of physiological and pathological functions. This protein's pivotal interfacial domains may be elucidated via its best experimentally proven blockers and molecular interaction software tools leading to drug discovery via a developed smart nanoparticle exhibiting a conjugated advanced re-engineered pharmacophore gating modulator exhibiting an improved potency and specificity experimentaly.

Method:

The Panx1 crystallized transmembrane protein's pivotal conformational domains, residues and atoms may be assayed towards elucidation and drug discovery via proven derivative blocker dynamic simulations (docking, protein-ligand interactions/dynamics, binding free energy/potential mean force, mutagenesis) coupled to experimental current analysis, dye uptake/efflux mutagenesis assays. Potential pharmacophore potency and specificity will be determined experimentally and computationally via IC_{50} Value determination, family channel affinity and free energy values respectively.

Results:

Computational Panx1-Blocker Derivative Complex generated results will elucidate the Panx1 gating regions, residues and atoms of interest via exhibiting mutagenesis binding free energy events and gating matching experimental mutagenesis gating current analysis and fluorescent dye molecular permeability data.

Conclusion:

The significance of this study if to elucidate the Panx1 blocking properties atomistically, this will catapult the development of a blocking re-engineered pharmacophore exhibiting unique and distinct dynamics towards controlling Panx1 gating to prevent diseased states. A deeper understanding of ion channel protein gating mechanisms will moreover advance biomedical engineering bettering human kind.

Goals, Objectives and Research Questions of the Study

The goal of this project is to elucidate the Pannexin-1 Cryo-EM mapped transmembrane protein's pivotal interfacial regions responsible for gating via best blocking partner derivative bindings precipitating allosteric (cryptic) and or steric blockage towards channel deactivation. The objective of this research is to develop a path towards the future development of pharmacophores exhibiting advanced channel regulation properties.

Via the below methods a channel structural and or steric closure will be exhibited answering the following questions:

What are the favored conformational domain(s) changes in time during ligand binding? (Method: Dynamic Dockings, Pymol.)

What is the functional potential of the developed partner; potency, specificity, selectivity? (Method: program predicting binding and functioning, experimental assays.)

How well characterized is the transmembrane interface(s) and binding partner? (Method: Dynamic Dockings, mutagenesis, current analysis/dye assays, binding free energy.)

What is the potency and efficacy of the partner during binding? (Method: in vitro/vivo developed assays.)

Which residues and atoms are pivotal towards channel blocking? (Method: Dynamic Dockings, mutagenesis, current analysis/dye assays, binding free energy.)

What is the best binding affinity and specificity? (Method: binding free energy and in vitro analysis.)

Methodology Overview

In silico molecular simulation places pivotal water molecules and ions within the final protein structure and precipitates a final energy minimized Panx1-ligand complex most resembling its natural state. Molecular Dynamic Simulations *in silico* may elucidate pivotal residue to inhibitor atomic interactions precipitating channel allosteric closure and or steric block via docking, binding free energy, mutagenesis and current analysis/dye assays.

Computational Tools:

GROMACS Molecular Dynamic Simulations elucidate Panx1-ligand atomic interactions via free energy, protein-ligand interactions and ligand dynamics.

Mutagenesis *in silico* may reproduce voltage clamp conditions measuring gating charge (current/dye analysis) via conformational change and steric block (i.e. single residue substitution precluding or enhancing substrate docking).

Binding Free Energy changes towards quantitative/qualitative drug prediction.

Potential Mean Force via an Umbrella Sampling Methodology determines sample improvement via ergodicity hindrance.

MedusaDock a flexible ligand receptor docking approach towards mutagenic elucidation.

Pymol/VMD interactive visualization softwares exhibiting structural/functional resolutions.